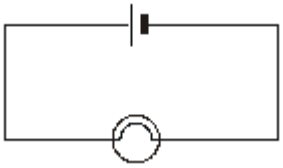
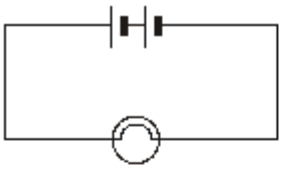
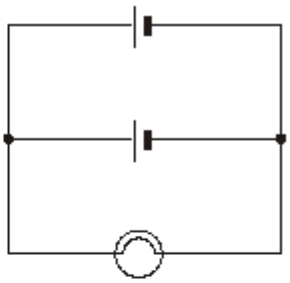
Q1. John connects up the circuit shown below.



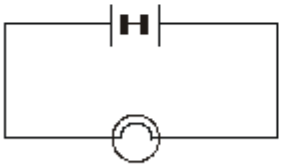
The bulb is not bright enough. His friend suggests four circuits which could be used to make the bulb brighter.



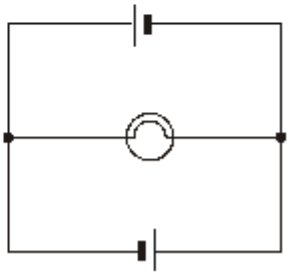
circuit A



circuit B



circuit D

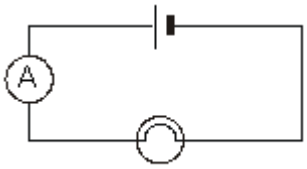


circuit C

(a) Which is the correct circuit to use: A, B, C or D? .....

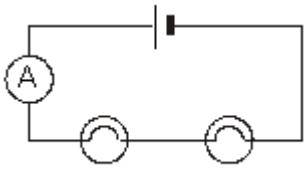
1 mark

Next John sets up circuit E and notes the reading on the ammeter.



circuit E

He then places another bulb in the circuit, to make circuit F. He notes the ammeter reading in circuit F.



circuit F

(b) How will the ammeter reading in circuit F compare with that in circuit E?

The reading in F is .....

1 mark

Explain you answer. .....

.....

1 mark

(c) Draw a circuit diagram in which **two** bulbs are lit as brightly as the bulb in circuit E, and the ammeter reading is the same as in circuit E.

1 mark

Maximum 4 marks

Q1.

(a) circuit A

if more than one letter is given award no mark

1

(b) smaller **or** lower

accept 'half'

1

any **one** from

- because the resistance is larger

  accept 'the resistance is smaller in E'

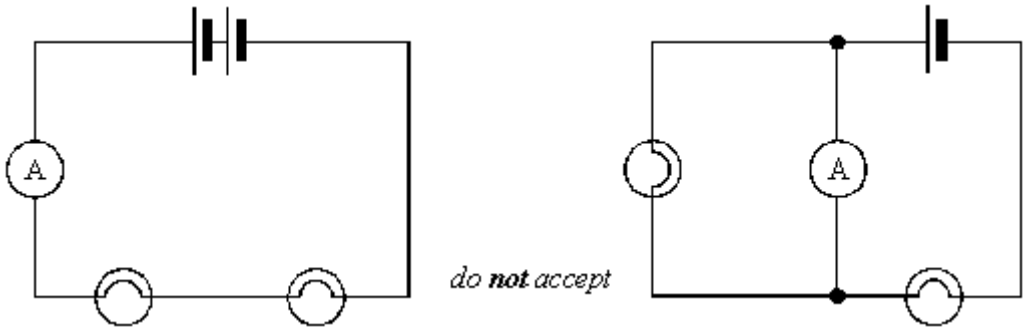
- because the current has to flow through more bulbs

  accept 'because there are more bulbs'

  *do **not** accept 'because two bulbs use up more energy'*
  ***or** 'because the current is less'*

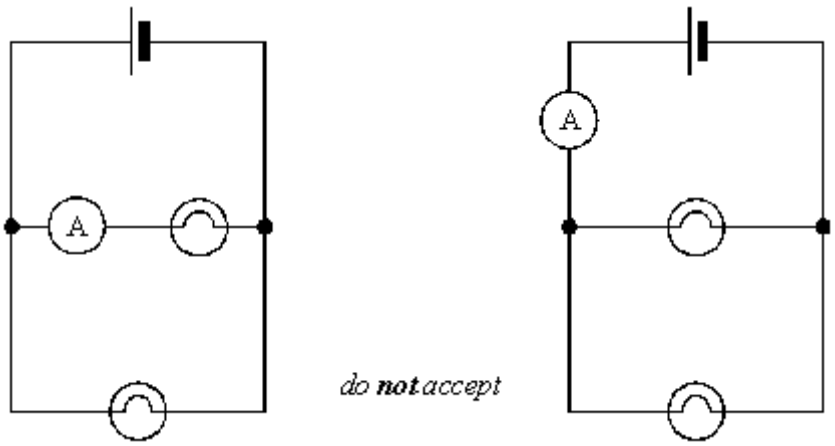
1

(c) If the two bulbs are in series, then two cells are needed.
The ammeter may be anywhere in the series circuit e.g.



*do **not** accept*

If the two bulbs are in parallel, then the ammeter must be in one of the branches of the
circuit e.g.



*do **not** accept*

1

[4]